CFD 2025-26 – Lab 12 – solution

The full code for this lab is provided in the notebook `CFDlab12.ipynb`.

Exercise 1

1 We introduce the following function spaces for the velocity, pressure, and level set functions:

$$V = \{\boldsymbol{v} \in [H^1_{\Gamma_{\mathrm{B}} \cup \Gamma_{\mathrm{T}}}(\Omega)]^2 \colon \boldsymbol{v} \cdot \boldsymbol{n} = 0 \text{ on } \Gamma_{\mathrm{L}} \cup \Gamma_{\mathrm{R}}\}, \qquad Q = L^2_0(\Omega), \qquad W = H^1(\Omega).$$

We point out that the pressure space Q needs a constraint on the average value (or on the value in a single point: cfr. previous labs) because pressure appears only through its gradient in the equations and BCs of the problem. By means of the usual integration by parts in the fluid problem, with the different boundary terms vanishing because all boundary conditions are homogeneous, we can derive the following weak formulations:

$$\begin{gathered}\text{Given } \boldsymbol{u}|_{t=0} = \boldsymbol{0}, \text{for any } t > 0 \text{ find } (\boldsymbol{u}, p) \in V \times Q \text{ s.t.}, \forall (\boldsymbol{v}, q) \in V \times Q, \\ \left(\rho \frac{\partial \boldsymbol{u}}{\partial t} + \rho (\boldsymbol{u} \cdot \nabla)\boldsymbol{u}, \boldsymbol{v}\right) + (2\mu\varepsilon(\boldsymbol{u}), \varepsilon(\boldsymbol{v})) - (p, \operatorname{div} \boldsymbol{v}) + (\operatorname{div} \boldsymbol{u}, q) = (\rho \boldsymbol{g}, \boldsymbol{v})\end{gathered} \tag{1}$$

$$\begin{gathered}\text{Given } \varphi|_{t=0} = \varphi^0, \text{for any } t > 0 \text{ find } \varphi \in W \text{ s.t.} \\ \left(\frac{\partial \varphi}{\partial t} + \boldsymbol{u} \cdot \nabla \varphi, \psi\right) = 0 \qquad \forall \psi \in W.\end{gathered} \tag{2}$$

2 Let X_h^r be the space of piecewise Lagrange polynomials of degree r over a triangulation of Ω. Introducing the spaces

$$V_h = V \cap X_h^{r_V}, \qquad Q_h = Q \cap X_h^{r_Q}, \qquad W_h = W \cap X_h^{r_W},$$

the finite element formulations of the two problems read as follows:

$$\begin{gathered}\text{Given } \boldsymbol{u}_h|_{t=0} = \boldsymbol{0}, \text{for any } t > 0 \text{ find } (\boldsymbol{u}_h, p_h) \in V_h \times Q_h \text{ s.t.}, \forall (\boldsymbol{v}_h, q_h) \in V \times Q, \\ \left(\rho_h \frac{\partial \boldsymbol{u}_h}{\partial t} + \rho_h (\boldsymbol{u}_h \cdot \nabla)\boldsymbol{u}_h, \boldsymbol{v}_h\right) + (2\mu\varepsilon(\boldsymbol{u}_h), \varepsilon(\boldsymbol{v}_h)) - (p_h, \operatorname{div} \boldsymbol{v}_h) + (\operatorname{div} \boldsymbol{u}_h, q_h) = (\rho_h \boldsymbol{g}, \boldsymbol{v}_h) \\ \text{Given } \varphi_h|_{t=0} \simeq \varphi^0, \text{for any } t > 0 \text{ find } \varphi_h \in W_h \text{ s.t.} \\ \left(\frac{\partial \varphi_h}{\partial t} + \boldsymbol{u}_h \cdot \nabla \varphi_h, \psi_h\right) = 0 \qquad \forall \psi_h \in W_h.\end{gathered}$$

where ρ_h implicitly depends on the triangulation, since it depends on φ_h:

$$\rho_h = \begin{cases} \rho_1, & \text{where } \varphi_h > 0, \\ \rho_2, & \text{elsewhere.} \end{cases}$$

3 - 4 Introducing a discretization $\{t^n = n\Delta t\}$ of time, the general fully discrete formulation of the two problems reads as follows, where one or more "$*$" indicate variables for which we can choose at which time we should evaluate them:

$$\text{Given } \boldsymbol{u}_h^0 = \boldsymbol{0}, \text{for any } n = 0,1,\dots \text{ find } (\boldsymbol{u}_h^{n+1}, p_h^{n+1}) \in V_h \times Q_h \text{ s.t.}$$
$$\left(\rho_h^* \frac{\boldsymbol{u}_h^{n+1}}{\Delta t} + \rho_h^* (\boldsymbol{u}_h^* \cdot \nabla)\boldsymbol{u}_h^{n+1}, \boldsymbol{v}_h\right) + (2\mu\varepsilon(\boldsymbol{u}_h^{n+1}), \varepsilon(\boldsymbol{v}_h)) - (p_h^{n+1}, \operatorname{div} \boldsymbol{v}_h) \tag{3}$$
$$+(\operatorname{div} \boldsymbol{u}_h^{n+1}, q_h) = \left(\rho_h^* \frac{\boldsymbol{u}_h^n}{\Delta t}, \boldsymbol{v}_h\right) + (\rho_h^* \boldsymbol{g}, \boldsymbol{v}_h) \qquad \forall (\boldsymbol{v}_h, q_h) \in V \times Q,$$

$$\text{Given } \varphi_h^0 \simeq \varphi^0, \text{for any } n = 0,1,\dots, \text{ find } \varphi_h^{n+1} \in W_h \text{ s.t.}$$
$$\left(\frac{\varphi_h^{n+1}}{\Delta t} + \boldsymbol{u}_h^{**} \cdot \nabla \varphi_h^{n+1}, \psi_h\right) = \left(\frac{\varphi_h^n}{\Delta t}, \psi_h\right) \qquad \forall \psi_h \in W_h. \tag{4}$$

The required Backward Euler scheme, with a semi-implicit treatment of the advection term in the fluid problem and an implicit treatment in the level-set problem, corresponds to the following choices:

$$\boldsymbol{u}_h^* = \boldsymbol{u}_h^n, \qquad \boldsymbol{u}_h^{**} = \boldsymbol{u}_h^{n+1}, \tag{5}$$

but still a choice has to be made on whether to determine the density ρ_h^* in terms of φ_h^n or φ_h^{n+1}. We may notice that the only nonlinearities remaining in the problems after the choices (5) are due to the coupling, since $\boldsymbol{u}_h^{n+1}$ appears as an advection field in the discrete level-set problem and φ_h^{n+1} may be chosen to compute the density. However, if we choose

$$\rho_h^* = \rho_h^n = \begin{cases} \rho_1, & \text{where } \varphi_h^n > 0, \\ \rho_2, & \text{elsewhere.} \end{cases} \tag{6}$$

we can adopt a so-called staggered approach and solve, at each time step, once the fluid problem and then once the level-set problem, without the need to resorting to iterative algorithms for the solution of the nonlinearities (e.g. fixed-point or Newton).

The overall structure of the solver is very similar to that of the lab on the coupled thermo-fluid problem: Navier-Stokes equations are coupled with another scalar system, and the numerical scheme is staggered. The main difference is that in this problem, the coupling passes through a piecewise-constant function, the fluid density ρ_h^{n+1}, defined in (6). As seen in the lab on SUPG stabilization, piece-wise constant functions can be defined by means of the function `conditional(CONDITION, VALUE_IF_TRUE, VALUE_IF_FALSE)`, with clear meaning of the input arguments. This choice is definitely suitable for the problem we consider here, in this regime. However, it is often advisable to introduce a smoothing of the density function, to improve stability and conservation properties in the numerical method for the fluid problem. This can be achieved by introducing a "smoothed Heaviside" function

$$H(s) = \frac{1}{2}\left(1 + \frac{s}{\sqrt{s^2 + h^2}}\right),$$

where h is the mesh element diameter, and then use

$$\widetilde{\rho}_h^{n+1} = \rho_1 H(\varphi_h^{n+1}) + \rho_2 H(1 - \varphi_h^{n+1}),$$

which smootly transitions from ρ_1 to ρ_2. In the code of `CFDlab12.ipynb`, this smoothing is toggled by the boolean variable `smoother_rho`: in the particular settings of this exercise,

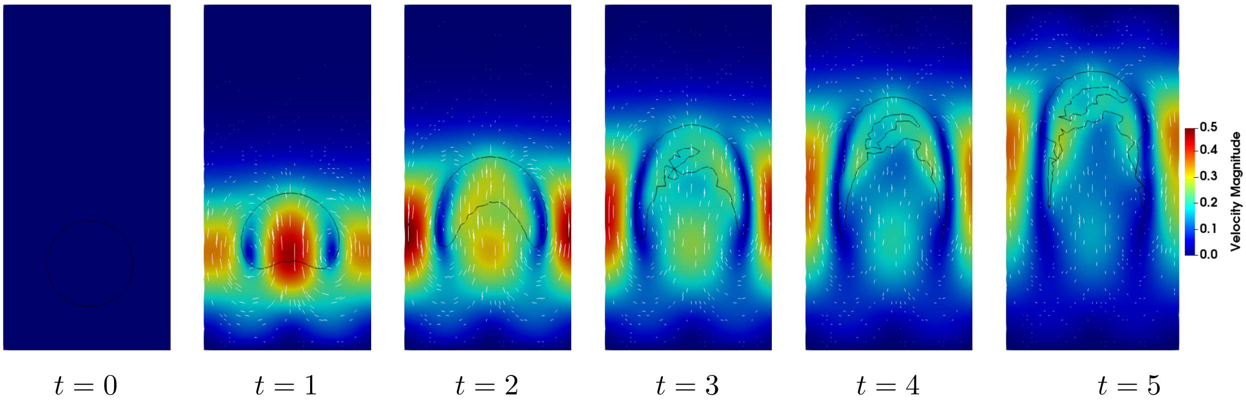

Figure 1: Velocity field and zero-level set $\{\boldsymbol{x} \colon \varphi_h(\boldsymbol{x}) = 0\}$ (black line) for different times.

the differences between the non-smoothed and smoothed case are negligible. Yet, this may not be the case in more complex applications.

The bubble interface, that is the level set $\{\boldsymbol{x} \colon \varphi_h(\boldsymbol{x}) = 0\}$, and the velocity field are reported in Figure 1. The bubble is driven upwards by buoyancy and two symmetric recirculations characterize the flow, following upwards the bubble in the center and going down along the lateral walls. This recirculation progressively pulls the bubble into a crescent shape. However, a region of heavier fluid is spuriously accumulating in the center of the bubble near t = 3, which progressively enlarges. Eventually, the solution has little physical sense (cf. t = 5).

Reducing the timestep does not improve significantly improve the results. This shows the limitation of a naive implementation of the level set method, which is better investigated in the next point.

5 The total mass can be computed as the integral of the density variable ρ_h^{n+1}, and we can do that for every simulated time. The resulting time evolution is reported in Figure 2. We may notice that the initial total mass of about 18.02 undergoes highly oscillating variations which significantly depart from the initial value, especially when the "inner bubble" of heavier fluid noticed in the first point is generated. A well-known reason for the unphysical behavior mentioned at the previous point is associated to the discrete level set function φ_h^n losing its property of being a distance function (in particular the property $|\nabla \varphi_h^n| = 1$) during the evolution of the system, due to numerical approximation. To restore such property, reinitialization of the level set function could be considered. A range of reinitialization strategies can be found in the literature for this problem.

Post-processing. The figures reported here can be reproduced using the Paraview state uploaded on WeBeep: in Paraview, go for *File* → *Load State* → select the file `lab12paraviewState.pvsm` you have downloaded → *Search files under specified directory* → select the directory in which your pvd/vtu files are.

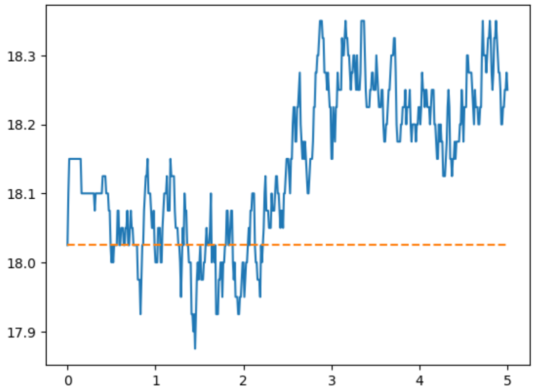

Figure 2: Evolution of total mass during the simulation. The red dashed line indicates the initial value.